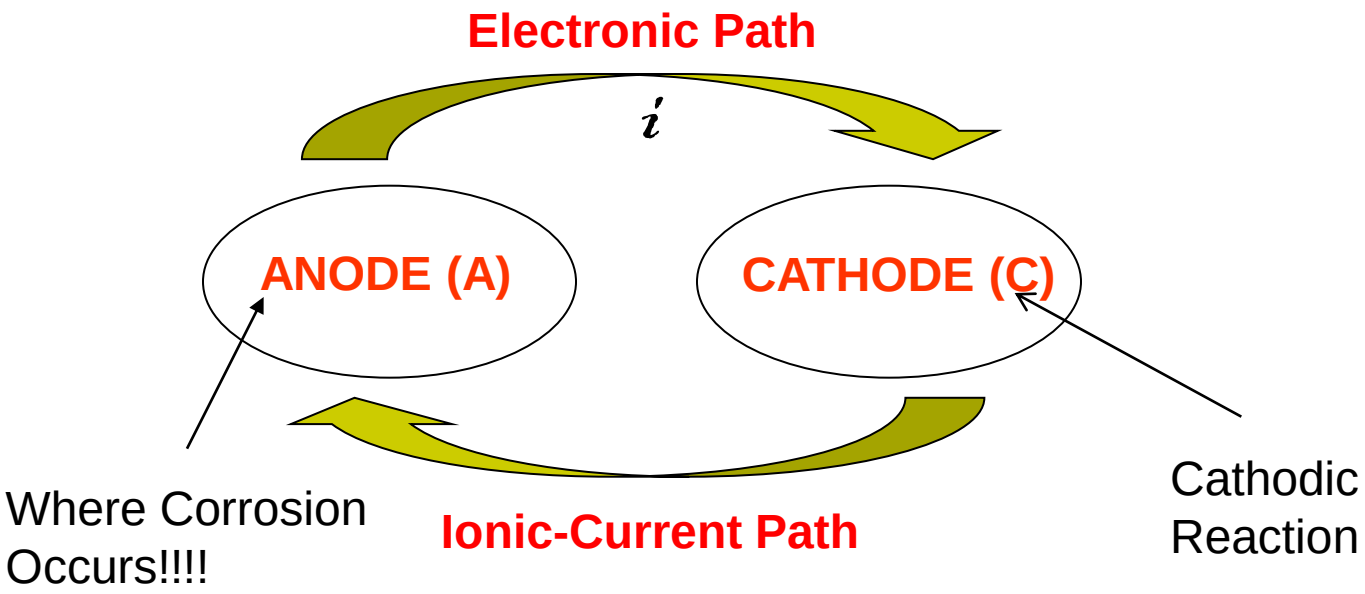
Chapter 5

Corrosion Protection

Remember

The five essential components of the corrosion cell (or Requirements for Electrochemical Corrosion)are:

1- Anodic Zones (A)
2- Cathodic Zones (C)
3- Electrical Contact Between Them
4- Ionically Conducting Solution
5- Cathodic Reactant (1+4 cathodic Reactions)



Metal or alloy

Anode + Cathode + Electronic Current Path

+

Corrosive Environment

Cathodic Reactant (1+4 !) + Ionic Current Path (moisture)

= ***Corrosion***

The Five Methods For Control Corrosion

- Material Selection
- Design
- Changing the Environment
- Changing the Metal Potential
- Surface Coating *(Zn, Sn, Plastic Paints etc)*

1- Corrosion Control by Material selection

(selection of proper material for a particular corrosive service depend on the environment)

- ✓Metallic [choose the proper metal or alloy]
- ✓Nonmetallic (use isolating joints and seals) [rubbers (natural and synthetic), plastics, ceramics, carbon and graphite, and wood]

Metals and Alloys

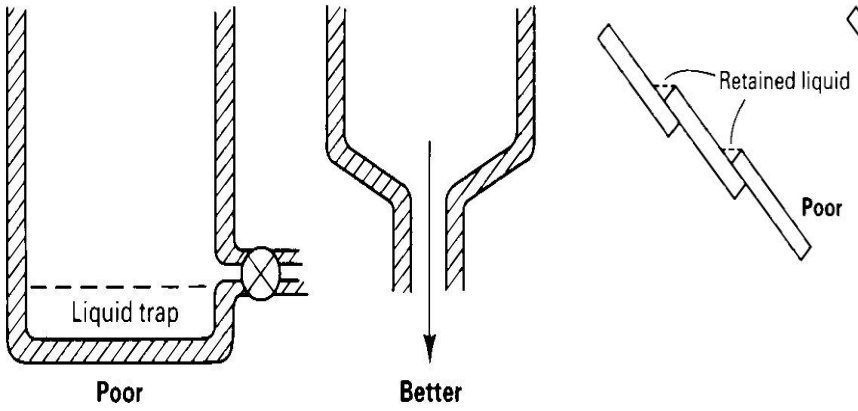
No	Environment	Proper material
1	Nitric acid	Stainless steels
2	Caustic	Nickel and nickel alloys
3	Hydrofluoric acid	Monel (Ni-Cu)
4	Hot hydrochloric acid	Hastelloys (Ni-Cr-Mo) (Chlorimets)
5	Dilute sulfuric acid	Lead
6	Nonstaining atmospheric exposure	Aluminum
7	Distilled water	Tin
8	Hot strong oxidizing solution	Titanium
9	Ultimate resistance	Tantalum
10	Concentrated sulfuric acid	Steel

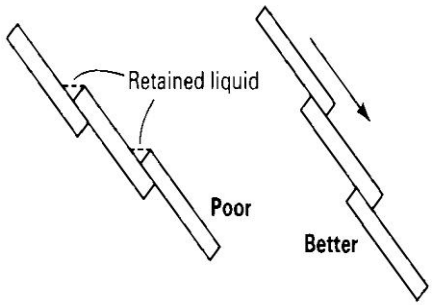
2- Control Corrosion by Design Modifications

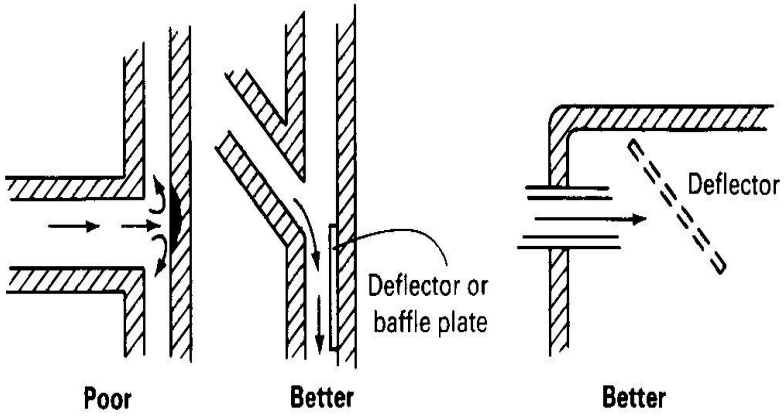
Corrosion Control must be properly addressed during the design, installation, and subsequent operation of systems.

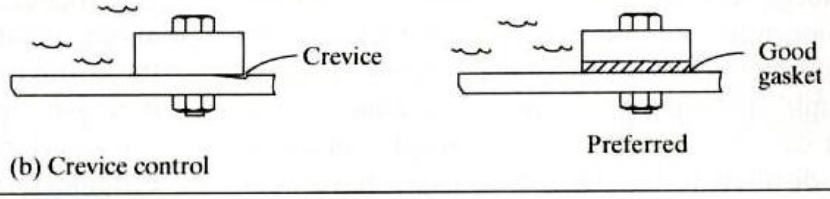
Design Do's & Don'ts

- To avoid SCC, you must avoid any excessive stresses.
- To avoid galvanic corrosion, you must (1) avoid any contact between dissimilar metals, (2) electrically isolate the two metals & (3) use high surface area anode.
- To avoid erosion corrosion, you must avoid any sharp bends in piping systems especially when high velocities and/or solid in suspension are involved.
- To avoid crevice/pitting corrosion, you must (1) avoid stagnant water/electrolyte (2) use good welding practice (weld rather than rivet) (3) use properly trimmed seals, rubbers, and gaskets.
- Avoid sharp corners where paint tends to be thinner and often starts to fail.
- Prevent solutions from drying/concentrating and ensure complete drainage of tanks to avoid any differential aeration (conc.) corrosion.
- Consider cathodic protections (1) zinc plating (galvanization), (2) use of sacrificial anodes of Zn or Mg on steel ship hull (3) use of impressed voltage from a battery.
- Design to exclude air – except for active-passive metals and alloys, they require O_2 for protective films.
- Most general rule : **AVOID ANY HETEROGENEITY** !!!

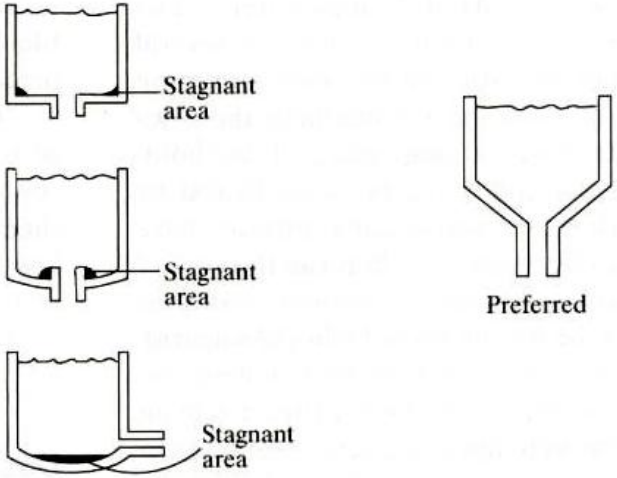




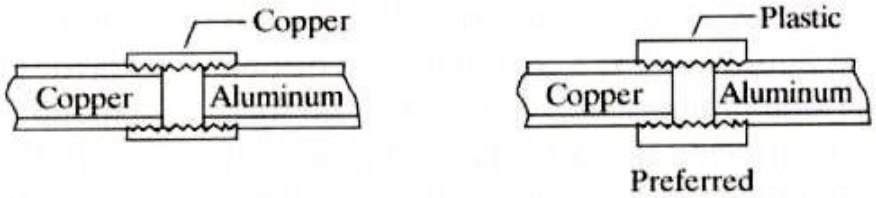




(b) Crevice control



(c) Provide proper drainage



(f) Prevent galvanic couples

3- Corrosion Control by Changing the Environment

- *Removal of O_2 from the system*
- *Control of pH (H^+ ions)*

Reactants of the cathodic reaction accompanying the corrosion process

- *Addition of complexing agents to remove (mask) the aggressive ions*
- *Addition of inhibitors to decrease the rate of oxidation and/or reduction reactions*

4- Control Corrosion by Changing the Metal Potential

Cathodic Protections

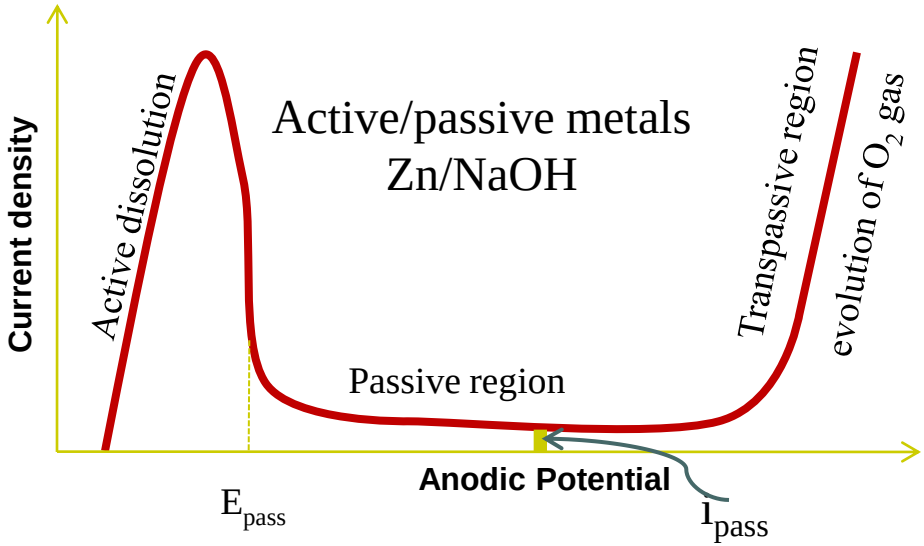
Shifting the metal potential to a more cathodic value. This shift in potential can be done by two different methods.

(a) Cathodic protection with impressed current

(b) Cathodic protection with sacrificial anodes

Anodic Protection

Shift metal potential into a range where passivation occurs. This is done by applying an external DC current to the anode (prior to its use) to form a protective passive film.



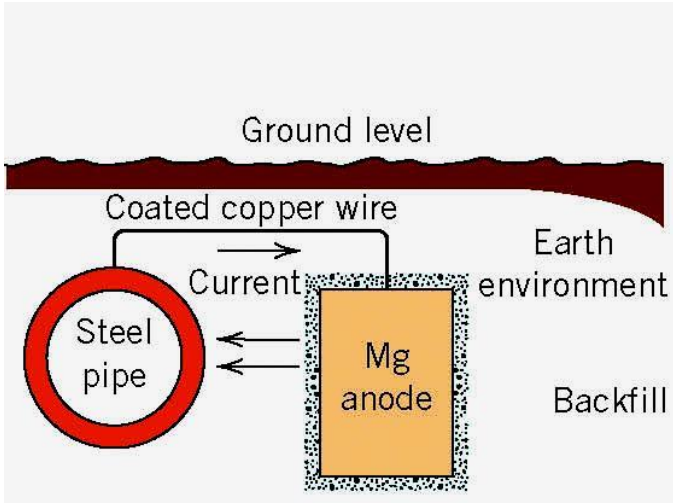
Cathodic Protection Methods

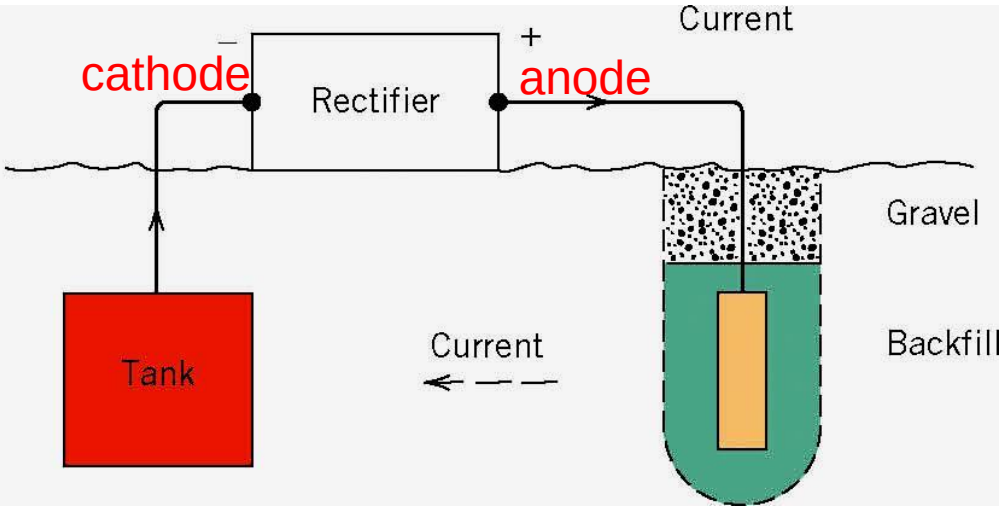
(a) Sacrificial Anode Method

Current is supplied by connecting the object to a **more reactive (easily corroded) metal** as a sacrificial anode. i.e. formation of galvanic cell in which the sacrificial anode is consumed (sacrificed).

(b) Impressed Current Method

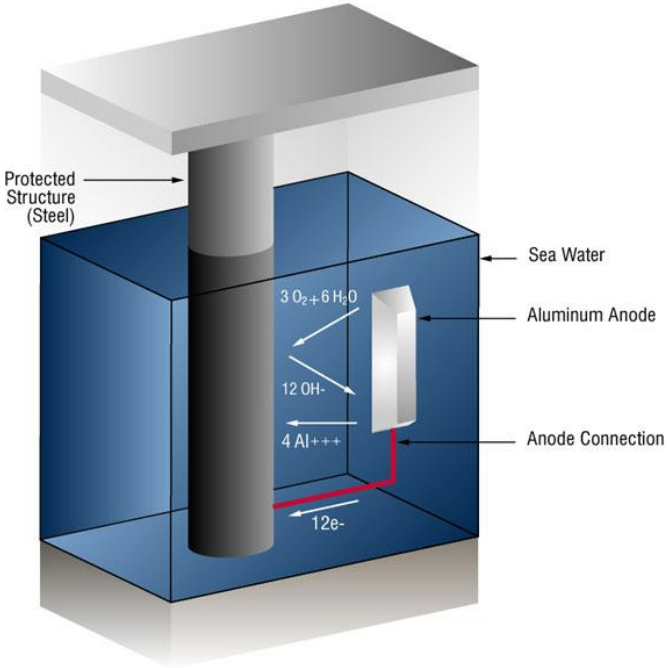
Here, an external current is applied from a DC source (e.g. a battery) and auxiliary anode. i.e. an electrolytic cell is formed where corrodible or inert auxiliary anode can be used





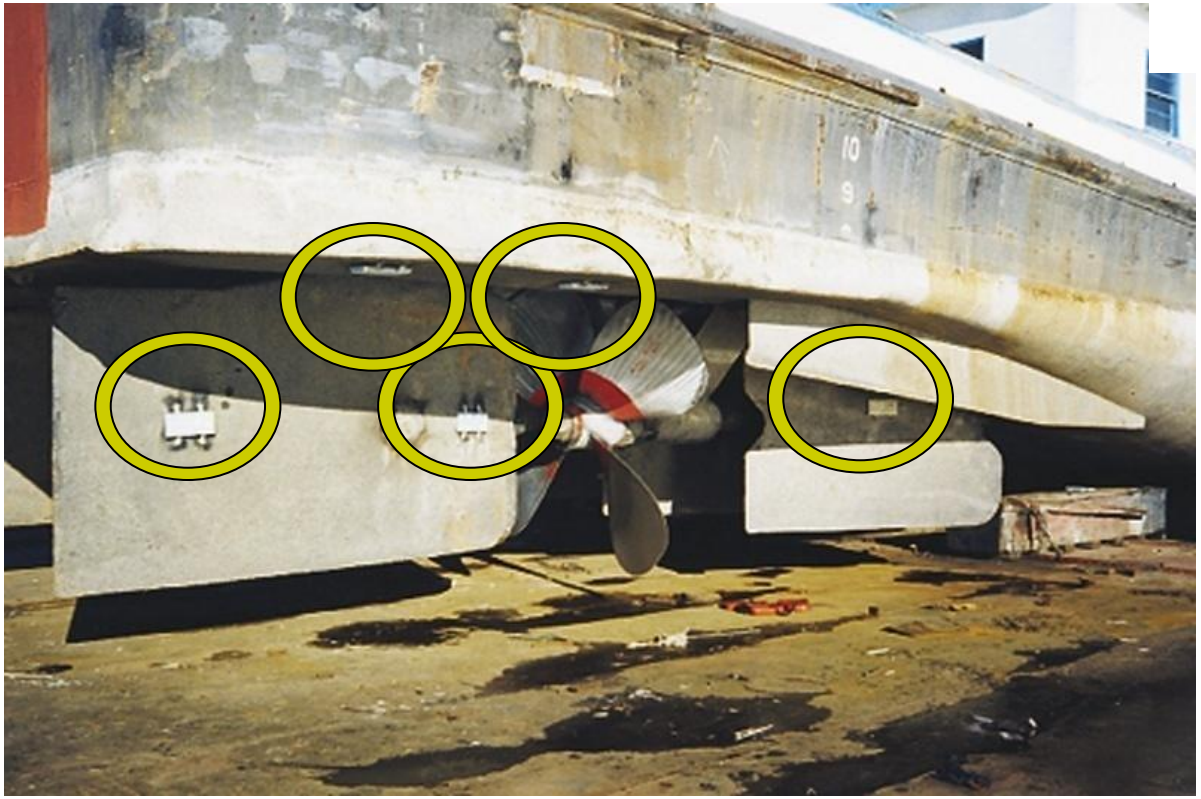
Cathodic Protection Methods

(a) Sacrificial Anode Method



Protection of Steel Structure

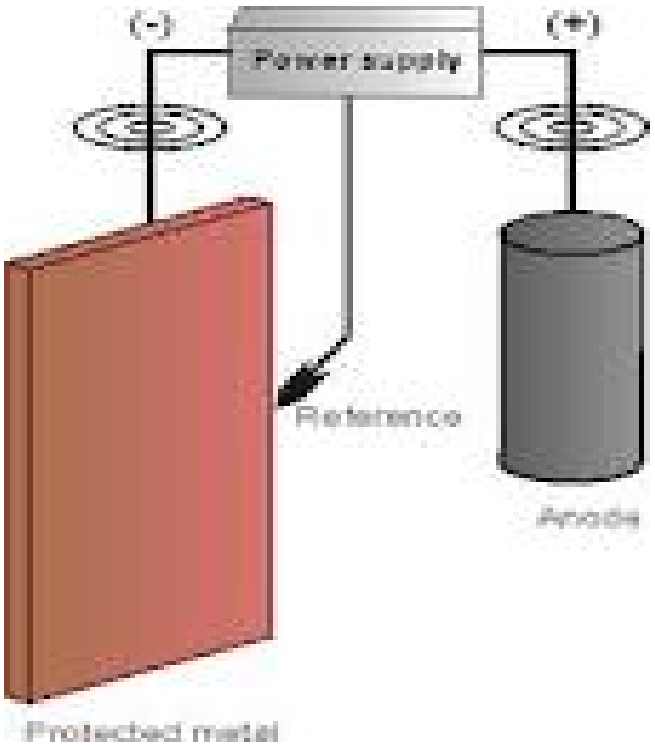
By connecting a piece of Zinc, Al or other anodic material to the steel, in this case, the zinc or other anodic metal is termed **"Sacrificed Anode"**.

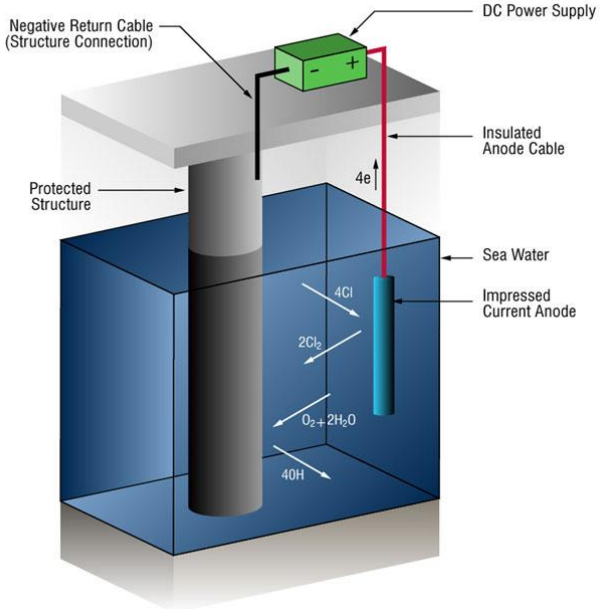


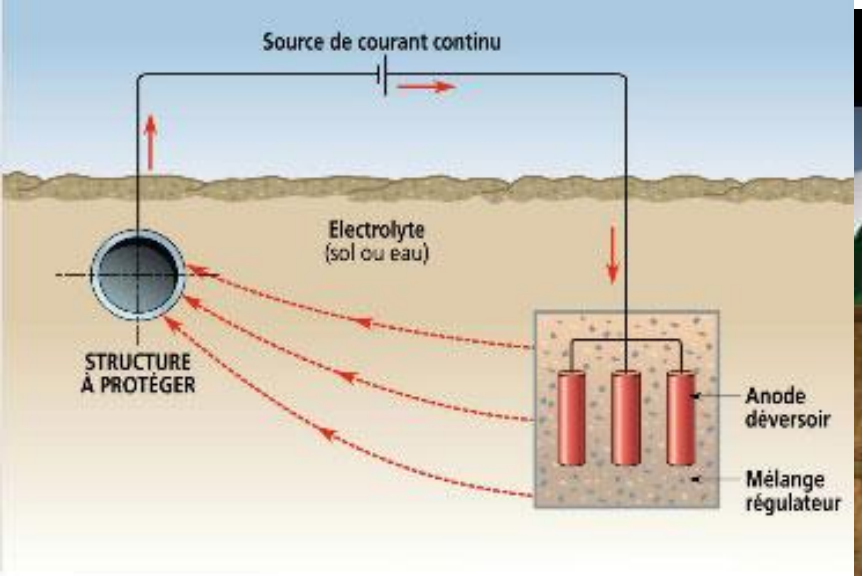
Cathodic protection of a ship hull using a zinc sacrificial anode

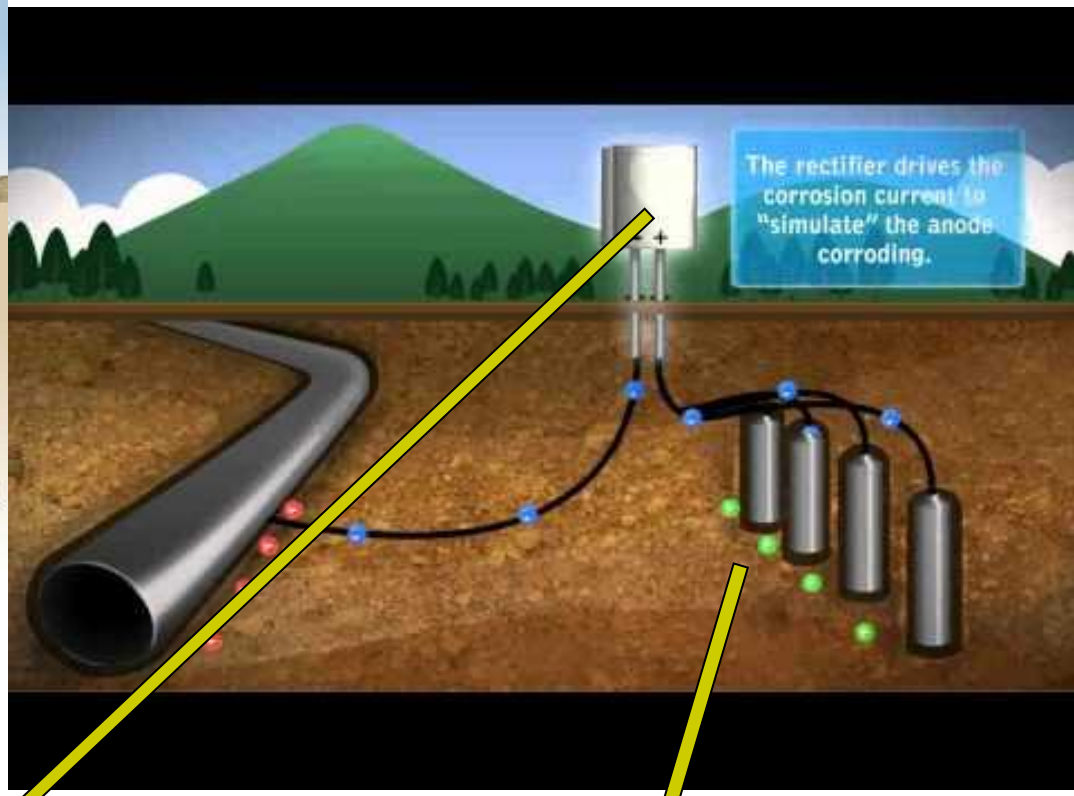
Cathodic Protection Methods

(b) Impressed current Method









Isolation des connexions indispensable
(sinon sortie de courant donc corrosion)

Système de protection par courant imposé

5- Corrosion Control by Surface Coatings

- Normally, it is the first line of defense against corrosion: provides barrier between a corrosive environment and the structure.
- No coating is perfect and usually contains holidays (voids) often resulting from improper application and/or damage.
- Coatings may act as sacrificial anode or release substance that inhibit corrosive attack on substrate.

Coatings

Metallic coatings
- Cathodic coatings Inert Barrier
- Anodic coatings Sacrificial Barrier

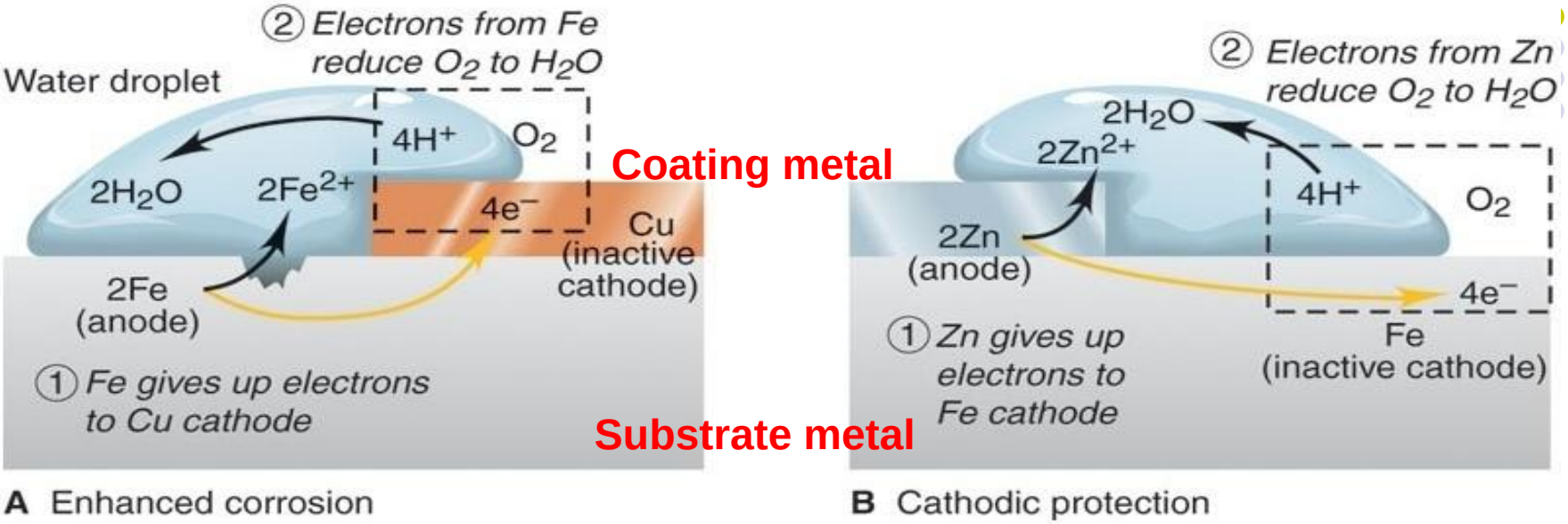
Non-metallic coatings
Inert Barrier

Inorganic coatings
- Conversion coating
- Portland cement coatings
- Vitreous enamels

Organic coatings
(Painting)

Metallic coatings:

(a) Cathodic coatings (Inert Barrier) **(b) Anodic coatings (Sacrificial Barrier)**



A Enhanced corrosion **B** Cathodic protection

The coating metal is:
more noble than the substrate metal.
i.e. **the coating metal is *cathodic* with respect to the substrate.**
e.g. **the coating of iron with silver, copper, nickel, Co, Sn, Pb on steel.**

This coating should be free of pores/discontinuity which creates small anode/cathode area leading to rapid enhanced attack at the damaged sites.

less noble than the substrate metal.
i.e. **the coating metal is *anodic* with respect to the substrate**
e.g. **the coating of Fe with Zn, Al.**

Exposed substrate will be cathodic & will be sacrificially protected.

Coating metal will form protective passive layer

Non-metallic coatings
Inorganic coatings

➢ Conversion coating

- In this process, the metal reacts with a compound so that thin insoluble coatings are formed on the metal surface. The surface is converted into the coating with a chemical or electro-chemical process.

(A) Phosphating (*Chemical Conversion*)

Phosphating is usually used on steel before painting.

(B) Chromating (*Chemical Conversion***) :**

(C) Oxide Coatings

The oxide treatments are done by heat, chemicals, or electrochemical reactions. **e.g. Anodizing (***electrochemical conversion* **).**

Properties to consider the quality of coating

- Average thickness of coating
- Porosity or continuity
- Adherence
- Uniformity of thickness
- Inert or passive